

Strengthening State Capacity: Civil Service Reform and Public Sector Performance during the Gilded Age[†]

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We use newly digitized records from the post office to study the effects of strengthened state capacity between 1875 and 1901. Exploiting the implementation of the Pendleton Act—a landmark statute that shielded bureaucrats from political interference—across US cities over two waves, we find that civil service reform reduced postal delivery errors and increased productivity. These improvements were most pronounced during election years when the reform dampened bureaucratic turnover. We provide suggestive evidence that reformed cities witnessed declining local partisan newspapers. Separating politics from administration, therefore, not only improved state effectiveness but also weakened the role of local politics. (JEL D24, D73, H83, J45, L82, L87, N41)

State capacity is a key driver of development and growth (Besley and Persson 2009). Civil service reforms are at the core of efforts to strengthen state capacity in developing and developed countries alike (World Bank 2000). While the exact nature of these reforms varies, a common objective is to create an impartial civil service (Besley et al. 2021; Evans and Rauch 1999). This objective is rooted in a long history of intellectual thought that considers a nonpartisan bureaucracy characterized by lifelong careers as an essential precondition for a well-functioning state (Weber 1922). Yet despite the apparent importance of civil service reforms, evidence on whether and how such reforms affect state effectiveness remains relatively scarce.

In this paper, we show that a strengthened bureaucracy has first-order impacts on the quality of public service delivery. We provide empirical evidence from America's watershed civil service reform—the implementation and expansion of the landmark

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1883 Pendleton Act within the US Post Office.¹ Prior to the Pendleton Act, the US federal government operated under the “spoils system,” a patronage-based structure that gave elected politicians and their respective political parties discretion over the hiring of bureaucrats. The spoils system allowed local parties to leverage federal appointments to amass substantial power, namely, through “the plunder of the salaries of the public servants for filling the party treasury ... the use of the power of removal for compelling those servants to be party henchmen ... and the use of the power of appointment for bribing and rewarding for party ends those not in the public service” (Eaton 1885, p. 18). As a result, corruption and powerful political machines dominated politics during the Gilded Age (1870–1900).

The Pendleton Act sought to reduce the influence of politics in administration and to reign in the power of political parties over the staffing and functioning of government. The statute replaced discretionary appointments with rule-based hiring, shielded bureaucrats from political interference, and abolished mandatory political contributions from bureaucrats (Hoogenboom 1959; Johnson and Libecap 1994). This seminal reform thus triggered a gradual transition of the federal administrative state toward a modern public organization that insulated bureaucrats from political forces (Theriault 2003), a move toward personnel practices commonly equated with “good” governance today (Evans and Rauch 1999, 2000). Despite the importance of this milestone reform, though, there is limited evidence for its effectiveness in improving bureaucratic performance and the quality of service delivery.

The public organization we examine, the post office, provides an ideal setting to study the impacts of civil service reform. First, in terms of personnel, the post office was, and remains today, the largest civilian agency within the federal government. Second, the post office was the principal method of long-distance communication in an expanding nation during the nineteenth century, thereby underscoring its economic importance. By providing the infrastructure needed to transmit personal correspondence and business information, the post office contributed to rapid economic growth during that time period (John 1995; Rogowski et al. 2021). Crucially for the purpose of being able to assess the impact of a nationwide reform, local post offices are present in virtually all parts of the country and perform the same functions everywhere, which is the collection and delivery of mail. This government service is carried out via a staff composed of carriers and clerks. The decentralized structure of the Post Office Department allows us to observe comparable measures of performance (e.g., delivery errors) and inputs (e.g., number of workers, personnel characteristics) across time and space. We combine these measures with the expansion of reform coverage to nearly 600 cities to identify the effects of the reform and the channels through which they operated.

Underpinning our study is a large-scale digitization of historical performance and personnel records of the US Post Office. We begin by using the Annual Reports of the United States Civil Service Commission to trace out the gradual expansion of the reform across nearly 600 cities.² We then link the timing of the civil service

¹ The post office was established in 1792, with President George Washington’s signing of the Postal Service Act. Today, the post office is known as the United States Postal Service, or USPS.

² We use the term “cities” interchangeably with the term “post office.” Post office subsidiaries within a city are referred to as branches.

reform to annual data on service quality and operations from two sources: the Annual Reports of the Postmaster General and the Annual Reports of the Superintendent of the Railway Mail Service. We examine the effects of two major civil service reform waves (1883 and 1893), which allows us to examine how this policy affected post office delivery errors across 1,763 cities from 1879 to 1901. We also examine detailed measures of cost efficiency and per unit productivity. To study how federal civil service reform affected the size and composition of the postal workforce, we also digitized individual-level data on the universe of clerks and carriers from the Official Registers of the United States (the “Official Registers” or “Registers”) for the period 1877–1901. This source contains rich data on the assignment, career progression, and background characteristics of postal workers. Finally, we use newspaper data to study how a depoliticized, more effective post office affected partisan politics and the strength of local parties—as measured by the circulation and presence of partisan newspapers.

One challenge to measuring the causal effect of civil service reforms is that such reforms are not randomly assigned but reflect deliberate policy choices (Ujhelyi 2014a). This raises the concern that the local adoption of civil service reform is correlated with city-specific factors that also affect postal performance. One advantage of our setting is that the reform’s rollout pattern was decided at the federal level. The reform initially covered larger cities before expanding to smaller urban areas. While the difference-in-differences approach helps address time-invariant sources of endogeneity, we also leverage institutional knowledge of the assignment rule. The first wave of the Pendleton Act of 1883 applied civil service reforms to cities with more than 50 postal workers. The later expansion via presidential order in 1893 was similarly size-based, depending on city population. These features allow us to augment our standard difference-in-differences by adopting a “control function” approach where we explicitly control for the size-based assignment variable in a time-varying fashion.

Our main finding is that civil service reform improved the quality of public service delivery. When relating postal delivery errors to the staggered expansion of the civil service reform, we find a significant and marked decrease in the number of delivery errors by 19–22 percent, consistent with an increase in the reliability of the postal service. This result holds when explicitly controlling for the assignment rule (based on post office staff size) and when using a large number of flexible, time-interacted controls to account for observable baseline differences between reformed and unreformed cities. Breaking down the reform effects by each of the two major reform waves of 1883 and 1893, we observe similar dynamic patterns with no pre-trends, providing evidence for both the external and internal validity of our estimates. We also show that the reform effects are not driven by any particular city or state and explicitly rule out confounders such as the concurrent improvement of complementary communications infrastructure or complementary local reforms. Finally, we corroborate our main finding based on delivery error data using a second set of productivity measures collected from an independent data source. Leveraging rich statistics from the free delivery service available for cities affected by the initial rollout of civil service protections under the 1883 Pendleton Act, we find that the civil service reform also led to significant productivity gains, as measured by the overall volume per carrier or the cost per volume.

Having documented substantial improvements in public service delivery driven by federal civil service reform, we leverage our granular data on inputs to study the underlying mechanisms. We first leverage the personnel records of clerks and carriers from the US Official Registers to rule out that the civil service reform increased the staff numbers in reformed cities. Turning next to an analysis of the selection margin, we use both individual traits reported in the Official Registers and additional background characteristics obtained by linking these personnel records to the complete count decennial censuses (“census”) to show that the reform did not significantly change the selection of personnel. We then use data on the overall mail volume to reject the possibility that the improved performance arose due to lower workload.

We end our discussion of mechanisms underlying the improved post office performance by conducting heterogeneity tests to shed light on the drivers of the observed performance gain. Under the spoils system, insecurity of tenure was a natural consequence of tying the fortunes of bureaucrats to elected politicians who often turned over (Hoogenboom 1959; Johnson and Libecap 1994). Using data on the careers of postal employees, we uncover substantial political cycles in exit among rank and file workers coinciding with presidential election years. These political cycles in turnover are dampened after the civil service reform, resulting in significant increases in the bureaucrats’ average duration of tenure. Using individual-level data, we find that the positive effects on retention apply to all bureaucrats serving in a reformed office, irrespective of their time of hire. Importantly, we find that the reduction in delivery errors and productivity improvements are largest during election years. Rather than improving the quality of service delivery by selecting higher-performing workers, civil service reform mainly improved bureaucrat performance by severing the ties between politicians and bureaucrats.

Finally, we consider whether civil service reform loosened the grip of partisan political actors at the local level. A major justification for reform of the civil service was to reduce the influence that local political parties had over the hiring of government employees (Mashaw 2009). Before the passage of the Pendleton Act, local political parties influenced the allocation of patronage posts and in turn relied on partisan loyalists appointed to government positions for both political and financial support. By reducing the availability of federal jobs that could be used to secure political support, civil service reform likely reduced the power of local parties. To test this hypothesis, we exploit the fact that newspapers during the Gilded Age were often partisan and backed by local political parties. This allows us to use the presence of explicitly partisan newspapers as a proxy for the strength of partisan politics. Indeed, the expansion of the Pendleton Act coincided with a marked increase in nonpartisan newspapers in the United States (Gentzkow et al. 2015), raising the question whether part of these stylized patterns can be attributed to the civil service reform. Using data on newspaper presence and circulation, we show that the civil service reform led to a decrease in partisan newspapers and an increase in the circulation of independent newspapers, consistent with a weakened role of local patronage politics. While suggestive, these findings underline the “double-dividend” of strengthened state capacity: civil service reform did not only result in a more effective bureaucracy but likely also helped reshape local politics in the United States.

Related Work.—Our findings contribute to several strands of literature. First, our work contributes to the literature on the impacts of state capacity (Besley and Persson 2009, 2010). While a larger body of work has documented long-run effects of state capacity across countries and regions (Dincecco and Katz 2016; Besley et al. 2021; Dell, Lane, and Querubin 2018), there is relatively less work on impact of efforts to weaken entrenched political actors, improve governance, and reduce patronage through civil service reform. On the subject of the effects of civil service reforms on policy, Ornaghi (2019) and Ujhelyi (2014b) show that civil service reforms can affect policy outcomes (using city and state reforms in the United States). Bostashvili and Ujhelyi (2019) show that state-level civil service reforms have the effect of stabilizing state government spending by weakening the political budget cycle. Finally, on the political implications of civil service reforms, Folke, Hirano, and Snyder (2011) find that state adoption of merit-based civil service systems weakened incumbent parties in the United States (as measured by state legislative wins). What differentiates our study from prior work is that we show the delinking of politics from public administration, effects on personnel, and effects on public service quality in a unified analysis. Moreover, the context—the first nationwide civil service reform in the United States—is an important setting.

Since policies that shape state capacity are often implemented across entire administrations or administrative units, a major challenge is the lack of counterfactuals. Existing work has relied on cross-country comparisons (Evans and Rauch 1999; Cornell, Knutsen, and Teorell 2020), as well as comparisons across a limited number of units, such as states or districts (Mehmood 2022; Liu et al. 2022; Moreira and Perez forthcoming; Mocanu 2022). The challenges arising from limited identifying variation are further exacerbated by the fact that bureaucratic performance is often difficult to measure, with many studies thus resorting to studying the reduced-form effects on downstream economic outcomes (Rauch 1995). Closest to our work is research by Moreira and Perez (forthcoming), who relate the rollout of the Pendleton Act in the Treasury’s Customs Service across 11 districts to revenue collection. Our study complements this work by focusing on the post office, which has long been one of the largest agencies of the federal government. The nationwide reach of the post office allows us to estimate the federal reform’s effects “at scale” (Muralidharan and Niehaus 2017) by leveraging the expansion of civil service reform from 23 to nearly 600 cities. Moreover, we can use a measure of public service delivery quality (delivery errors) that is readily comparable across affected and unaffected cities.

Second, we contribute to the literature on the personnel economics of the state. There is now a large body of work that studies how policies aimed at introducing merit-based personnel practices affect performance (Dal Bó, Finan, and Rossi 2013; Finan, Olken, and Pande 2015). Much of this work has shown the effectiveness of incentives (Khan, Khwaja, and Olken 2015, 2019; Deserranno, Kastrau, and León-Ciliotta 2021), demonstrated how opaque links between effort and reward weaken performance (Xu 2018; Bertrand et al. 2019; de Janvry et al. 2023), or focused on the role of bureaucratic selection (Moreira and Perez forthcoming; Mocanu 2022; Weaver 2021; Voth and Xu 2022; Dahis, Schiavon, and Scot 2023; Mehmood 2022; Riaño 2022). By leveraging both city-level and individual-level data, we connect the “macro” literature on civil service reform to this fast-growing micro-level literature.

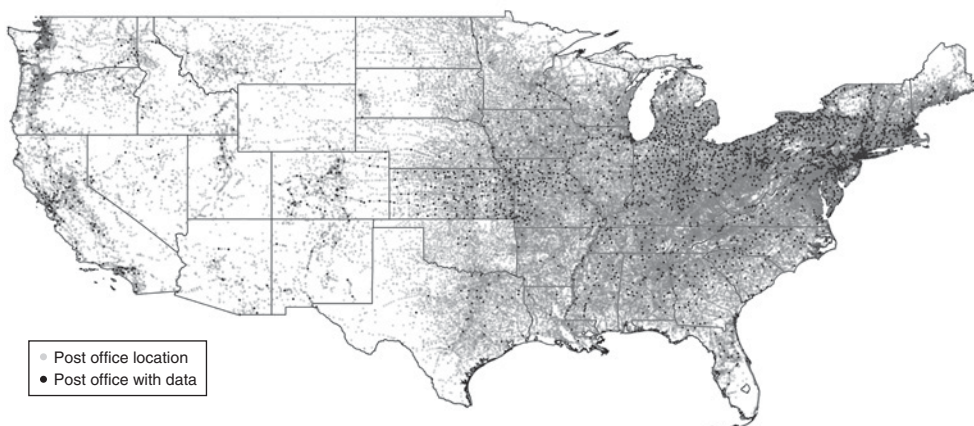
In contrast to Moreira and Perez (forthcoming), who focus primarily on the introduction of competitive examinations as the key margin for affecting bureaucrat performance, our result suggests that the Pendleton Act's measurable influence on performance worked through creating a civil service that is shielded from political interference. By providing evidence that the civil service reform improved performance by creating stable careers through the insulation of bureaucrats from electoral cycles, our results resonate with the literature on the costs of political turnover (Iyer and Mani 2012; Akhtari, Moreira, and Trucco 2022; Colonnelli, Prem, and Teso 2020).

Finally, we contribute to research on American economic history in a few key ways. Most directly, we advance the literature on the evolution of the American government during an important period of political and economic development. Existing studies primarily assess the political economy aspects of state- and city-level civil service reforms at the turn of the nineteenth century (Folke, Hirano, and Snyder 2011) as well as the impact of such changes on policy and performance outcomes (Ujhelyi 2014b; Ornaghi 2019). We build on this literature as well as on the work of Moreira and Perez (forthcoming, 2022), by measuring the impact of the nation's signature set of *federal* civil service reforms on performance within the post office. We also link our findings of improved bureaucratic performance to measures of economic performance in the spirit of Evans and Rauch (1999), thus suggesting that state investments in depoliticized and efficient public service delivery have downstream consequences. By showing that increased newspaper circulation is driven by nonpartisan rather than partisan sources, we contribute to research on the source of declining media partisanship at the end of the nineteenth century (Gentzkow, Glaeser, and Goldin 2006; Petrova 2011; Hirano and Snyder 2020). Lastly, we contribute to research on the role and structure of the post office in American economic history (Boustan and Margo 2009; Acemoglu, Moscona, and Robinson 2016).

I. Historical Context

US Post Office.—Post offices have been a key ingredient for state building in several countries (Rogowski et al. 2021; Geloso and Makovi 2020; Andrabia 2020; Chong et al. 2014; Maclachlan 2011). The US Post Office was created by the Post Office Act of 1792 and has been viewed by some as a key institutional driver of growth and social progress. During its early years, the post office facilitated communication and the spread of information between distant areas of the nation and contributed to the development of an informed citizenry (John 1995). Indeed, Alexis de Tocqueville described the American postal system as a “great link between minds” and admired how mail traveled from eastern cities like New York and Philadelphia to (then) sparsely populated areas, like Detroit, located on America's western frontier (Tocqueville 1969). The expansion of mail service also facilitated economic transactions across greater distances, allowing sellers of goods to advertise their products using catalogs and to complete sales via mail orders. Roper (1917), for instance, describes how money and goods “pass[ed] through the countless postal channels.” Relatedly, the post office provided a system of banking and allowed investors to find opportunities. Business-related mail

Panel A. All post offices



Panel B. Expansion of civil service reform

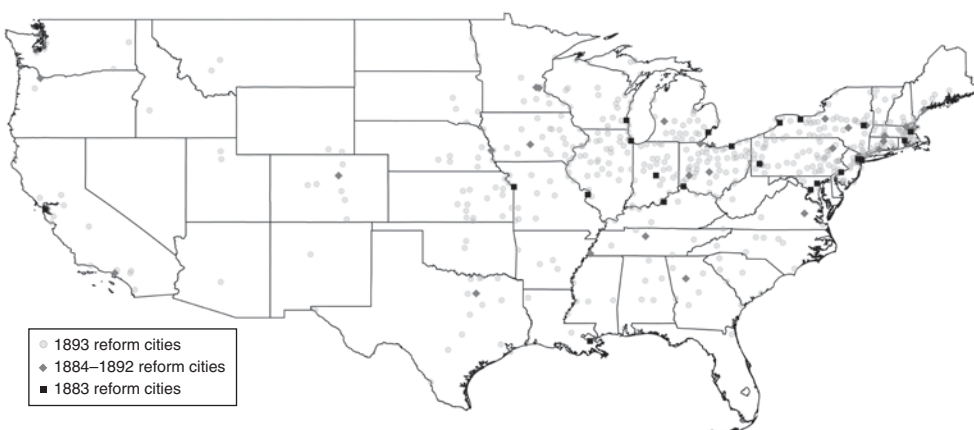


FIGURE 1. SPATIAL DISTRIBUTION OF POSTAL OFFICES AND EXPANSION OF THE CIVIL SERVICE REFORM

Notes: Panel A shows the locations of all post offices open between 1860 and 1905, as well as the locations for which performance data on delivery error rates or free delivery outcomes are available. Panel B shows the locations of reformed post offices and their timing. There are 23 locations that are reformed in 1883, 30 locations between 1884 and 1892, and 556 locations in 1893. The shapefiles are from National Weather Service (2023).

comprised a substantial portion of total correspondence in the nineteenth century, leading some to conclude that America’s “commitment to postal service formed ... the foundation for [nationwide] growth” (Henkin 2007).

One can see the potential for the post office to transform the nation economically, politically, and socially by observing its reach after just a century of existence. Figure 1, panel A shows the locations of all post offices that were open between 1860 and 1905 (Blevins 2021) and highlights the ubiquity of this aspect of US state capacity. The presence of post offices in all corners of the nation underscores the widely held view that in the nineteenth century, “the postal system was *the* central government” (John 1995) (emphasis added). Panel B highlights the subset of cities whose post offices are ultimately affected by civil service reform.

The Patronage Era of the American Government.—The US government grew substantially during the nineteenth century in response to factors such as westward expansion, commercial development, and the demands of the Civil War. The post office was the nation's largest federal agency—"the mighty arm of civil government," according to the *New York Times*. At the onset of the Pendleton Act, postal workers alone made up 34 percent of the entire federal workforce, larger in terms of labor force than any other department (US Census 1880). Moreover, given the importance of the post office to economic activity, the personnel responsible for this central communications infrastructure had the potential to shape American growth.

Despite its role in facilitating communication and economic activity, the early post office was rife with corruption and inefficiencies (Carpenter 2001). One likely source of organizational ineffectiveness was the "spoils system," a system of patronage-based hiring and firing that pervaded much of the federal bureaucracy. For much of the nineteenth century, many federal employees were hired at least partially based on political considerations (Johnson and Libecap 1994). Politically motivated personnel decisions were commonplace in many post offices, which hired large numbers of staff. Under the spoils system, patronage employees were expected to be politically active on behalf of their benefactors by assisting in political campaign activities and contributing part of their salaries to politicians in the form of "political assessments." Because bureaucrats were tied to particular politicians or parties, high levels of bureaucratic turnover were also commonplace during this period (Fowler 1943). In New York City, for example, the dominance of partisan considerations led to "short terms of office and easy" as well as "frequent removals" (Eaton 1910).

The spoils system, which included high levels of politically motivated employee turnover, imposed significant costs in terms of both the quality and efficiency of service provision. Bureaucrat ineptitude caused in part by a corrupt patronage system hampered the quality of postal service. Johnson and Libecap (1994), for instance, describe the anger of political constituents about the impact of patronage on public services during the years preceding federal reform: "[s]candals and charges of fraud and inefficiency were linked to the country's largest post offices and customhouses. Patronage was blamed, and the president and members of Congress with major federal facilities in their districts were increasingly pressured to reform the system" (p. 20). Newspapers, which comprised a substantial fraction of mail volume, complained of the "long-enduring irregularity of the mails, and the excessive carelessness or gross ignorance ... [within] a good many of the post offices" (Foley 1997).

Problems were particularly acute in cities, where patronage politics were salient. In the New York City Post Office, for instance, one report described the "incompetency, neglect, confusion and drunkenness" of postal staff. Indeed, one incoming postmaster prior to the early 1880s stumbled upon many bags of undelivered mail that were scattered throughout the post office building (Hoogenboom 1959). By the mid-1870s, even members of Congress acknowledged that the quality of government service provision was often "poor at best" and that the administrative state required reform (Johnson and Libecap 1994).

The Pendleton Act.—In 1883, Congress passed the Civil Service Reform Act—better known as the Pendleton Act—which aimed to limit the influence of politics over public administration of the laws through a bundle of measures. The law created

a merit-based civil service by replacing hiring based on discretion with rule-based personnel decisions relying on competitive exams and performance. The Pendleton Act also provided workers with protection from political removal, as they were vulnerable to being fired summarily during periods of political turnover (Masur 2013). Finally, it outlawed for all federal workers the use of “assessments,” fees that civil servants were asked to pay in return for their appointment.³

The passing of the Act was a culmination of pro-reform sentiment that had been developing since the end of the Civil War. Federal legislation was ultimately catalyzed by the assassination of President Garfield by a disappointed office seeker in 1881 (Theriault 2003). Under the initial 1883 statute, the government immediately “classified” around 10 percent of federal employees as formal members of the protected civil service (Johnson and Libecap 1994). Civil service protections applied initially to the federal workforce in DC, as well as to large customhouses and post offices across the country. Among these departments, the post office was by far the largest affected department in terms of affected manpower, making up a third of the total federal workforce.⁴ The statute was designed such that the post office was among the first departments to be affected by civil service reform. The targeting of the post office was significant since the provision of postal services was an important obligation of the federal government in the nineteenth century (Johnson and Libecap 1994).

Candidates for classified jobs were selected through open, competitive examinations and protected from arbitrary firing. During the first two decades of the reform era, postal employees were classified as protected civil servants based on the city in which they worked (Hoogenboom 1959). Classified post offices were initially restricted to post offices with at least 50 employees, meaning that most of the positions classified were located in urban post offices.⁵ According to the Civil Service Commission, this threshold choice was guided by both an attempt to experiment and target large post offices where the negative consequences of patronage were deemed more serious (United States Civil Service Commission 1883). Over subsequent years, though, additional local post offices became “classified” for purposes of civil service protections. By 1921, the proportion of the federal civilian workforce covered by the civil service system had grown to 80 percent. Importantly, the law also delegated to the president the authority to issue executive orders moving additional jobs from patronage into the new civil service system. In 1893, President Benjamin Harrison issued an executive order that classified all post offices with free delivery service as civil service–protected (United States Civil Service Commission 1893).⁶ Harrison’s order extended civil service reform to 556 additional cities, expanding the civil service reform well beyond the biggest 23 large cities that were covered in the initial 1883 reform wave.

³The ban on assessment fees applied evenly across the entire government when the Pendleton Act was implemented. The effects we document using our difference-in-differences design, therefore, occur in addition to any effect caused by the ban on assessment fees, which we are unable to separately identify in this setting.

⁴In comparison, the customs office—the second unit covered by the Pendleton Act—made up 3 percent of the federal workforce (estimate based on census and Official Registers data for 1880 and 1881, respectively).

⁵Unlike the customs (compare Moreira and Perez forthcoming), there were no exemptions, and the classification applied to all clerks and carriers.

⁶We describe free postal delivery service in more detail in Section II. Under the original statute, the president was authorized to extend the provisions of the Pendleton Act to other post offices and also to other branches of the service (Lyman 1893).

II. Data and Descriptive Statistics

A. Data Sources

We combine multiple sources of administrative and personnel data to construct our main dataset. Here, we briefly describe the main sources of the data.

Government Performance Measures.—Our performance measures come from two sources of administrative data that were compiled regularly by the US Post Office during the nineteenth century: the Annual Report of the General Superintendent of the Railway Mail Service (Railway Mail Service 1883) and the Annual Report of the First Assistant Postmaster General (United States Post Office Department 1883). Our main measure of post office performance captures errors in delivery and was recorded in a series of annual post office statistics, the *Statement of errors in the distribution and forwarding of mail*. These statistics were systematically recorded in the Annual Report of the General Superintendent of the Railway Mail Service and were frequently used to benchmark the performance of post offices.

The Railway Mail Service (RMS) was primarily responsible for intercity mail delivery during the nineteenth century and first half of the twentieth century. The delivery of mail across cities relied on the accurate collection and routing of mail by clerks working in local post offices. Postal clerks would collect and group together all articles of mail (letters, newspapers, etc.) that were intended for a particular destination. To facilitate sorting and transport by the RMS, the *Postal Laws and Regulation* required clerks to place a label, called a “facing slip” on top of a packet of mail items that indicated the destination and the name of the processing clerk in the origin city (United States Post Office 1879). These slips not only facilitated a more efficient distribution process but also created a record of all errors or irregularities made by postal clerks in sending cities.

Mistakes were identified by clerks of the RMS, who, in the process of sorting mail for intercity distribution, examined all items within a bundle for errors in where mail was directed. Recorded errors were then double-checked by another expert examiner (White 1910). Records of mistakes by either sending clerks or RMS clerks were then submitted to supervisory postal employees to track employee performance. These errors recorded on facing slips thus became a way city post offices were monitored.

The historical statistics report two measures of delivery errors at the city-year level: an extensive margin measure capturing the number of slips that contained errors as well as a measure that captures both the extensive and intensive margin—the total number of errors on returned slips. We focus on the latter measure, the total number of errors on returned slips, which we subsequently refer to as “delivery errors.” This measure allows us to capture variation at both the extensive and intensive margin, providing an accurate representation of the total number of mistakes made by origin post offices.⁷ Finally, a key advantage of this performance measure is that the RMS was an independent arm of the US Post Office. As such, it was not associated with a particular

⁷The correlation coefficient between both measures of errors—the number of incorrect slips returned and the total number of errors on returned slips—is 0.97. Our results hold using both measures from the General Superintendent’s Annual Reports.

post office or subject to the city-specific rollout of the civil service reform, alleviating concerns over strategic misreporting. A natural limitation of this measure is that it only captures errors in the delivery of mail sent across different cities, which is likely to understate the total number of delivery errors. We collect the annual statements for the period 1879–1901, covering a total of 2,319 cities in our sample.⁸

The second key source of information on post office performance that we use comes from another administrative source, the Annual Report of the First Assistant Postmaster General. The First Assistant Postmaster General (FAPG) was the deputy responsible for overseeing the post office's City Free Delivery service, a departmental innovation that allowed letter carriers to deliver mail to a recipient's home within certain cities. The FAPG's annual report provided a statistical appendix relating to the operation and performance of the post office's free delivery service. These data provide a unique portrait of the local free delivery operation, allowing us to measure the number of carriers in service, the amount of mail delivered and collected, the number of pieces handled, the cost of service, and the total amount of postage each year. To our knowledge, there has been no systematic examination of the postal service using these administrative data.⁹ We use these secondary measures to complement our main performance measure and explore the mechanisms through which impacts (if any) occur. A limitation is that these series are only reported up to 1891, and we digitize the annual statements for the period 1875–1891. This covers a total of 512 large cities with free delivery service. Figure 1, panel A summarizes our data availability. As the figure shows, we cover all urban areas across the nation.

Personnel Records.—To document how the postal reform affects the composition of the personnel and their career progression, we digitized the series of the Official Registers of the United States, Part II (“Official Registers” or “Registers”). Issued biennially, the Official Registers listed every employee of the US federal government for the time period 1816–1921, and high-ranking federal officials thereafter. Given the large size of the US Post Office in terms of workforce, the personnel listing of the post office was published separately as Part II of the publication series. We digitized personnel data on all clerks and letter carriers for the time period 1877–1901, covering our main sample period. Clerks and letter carriers make up the personnel of each city's post office and are thus the main occupations that were targeted by the postal reform. These are also occupations for which the Registers record detailed information about the names, birth states, work locations, and salaries. The availability of birth state and work locations, in particular, is crucial to allow us to link our data to the full-count decennial census to obtain additional background characteristics. Overall, we digitized a total of 3,108 pages, corresponding to a total of 297,932 individual-year observations.

Data on Newspaper Partisanship.—To measure the impact of civil service reform on local outcomes, we examine newspaper partisanship and circulation. This choice

⁸Unfortunately, we were unable to find annual statements going back earlier than 1879. This limits the amount of pre-period we can examine for the 1883 reform.

⁹Carpenter (2000) uses data on *rural* free delivery from three states to examine aggregate trends in the cost of postal service during 1890–1915. Our study studies postal performance across the entire nation from 1875 to 1901.

is guided by the historical literature, which documents a significant rise in independent newspapers during our study period (Gentzkow, Glaeser, and Goldin 2006; Petrova 2011; Hirano and Snyder 2020). We use data from Gentzkow, Shapiro, and Sinkinson (2011) to obtain the number of partisan and independent newspapers with their circulation at the city level for each presidential election. The underlying data come from digitized historical newspaper directories (*Rowell's and Ayer's American Newspaper Directories*) that were used by advertisers. Our final sample covers 1,037 cities for all election years between 1879 and 1900.

B. Empirical Patterns of Civil Service Reform Rollout

Reform Rollout.—The Pendleton Act was designed to shield civil servants from political interference. Under the statute, employees in classified positions would be selected through open, competitive examinations and would also be protected from firing without cause (United States Civil Service Commission 1883). As described above, the original statute affected postal jobs in a subset of cities, and these protections were subsequently rolled out to other places. To determine the timing of federal civil service rules across cities, we use the Annual Reports of the United States Civil Service Commission. First issued in 1883, these annual volumes document the progress of civil service reform throughout the nation. They report information on the coverage of civil service protections across locations, positions, and departments, as well as statistics on the number of examinations, the number of appointments, and aggregate statistics on candidates selected for federal jobs. Importantly for our purposes, these reports include lists of “classified” locations where formal civil service rules applied, thus allowing us to track the expansion of civil service reform within the Post Office Department over time. We use this information to create a city-year panel of civil service coverage within American post offices.

We begin our analysis by examining the determinants of civil service reform across the nation. Figure 1, panel B displays the rollout of the reform during our study period. The rollout of the civil service reform was uneven across space but touched most parts of the nation. Reform occurred in two primary waves: the initial wave of the 1883 statute, which brought reform to 23 large cities, and the wave of 1893, which saw an expansion of civil service coverage to all then-unclassified 556 post offices that offered free delivery services. During the period between these two expansion waves, 30 cities became classified as they crossed the 50 employees threshold of the 1883 act. In our main analysis, we focus exclusively on the 1883 and 1893 reform waves. Unlike cities that are gradually reformed as they grow past the designated 50-employee threshold between 1884 and 1892, assignment to treatment for the two major waves left less room for endogenous anticipatory responses.¹⁰

To formally assess the differential pattern of the rollout, Table 1 compares the characteristics of cities targeted by the reform to those not targeted. Panel A reports city-level baseline characteristics, and panel B reports county-level characteristics. Given the size-based rollout of the Pendleton Act, reformed cities differ from unreformed ones in a variety of ways. For example, cities with post offices that became

¹⁰In later robustness checks, we also show that our results are comparable when including all reform waves

TABLE 1—DESCRIPTIVE STATISTICS OF REFORMED AND UNREFORMED CITIES

	Mean control (1)	Difference treatment-control in reform wave			
		1883 (2)	1884–1892 (3)	1893 (4)	Pooled (5)
<i>Panel A. Post office level</i>					
Post office staff	3.38	170.144 (51.125)	18.964 (1.698)	3.923 (0.230)	15.483 (3.731)
log(city population)	8.39	3.320 (0.176)	2.270 (0.091)	1.646 (0.037)	1.793 (0.039)
log(delivery errors)	4.41	4.286 (0.261)	2.787 (0.181)	1.758 (0.084)	2.021 (0.083)
Year established	1837.06	−33.598 (8.622)	−11.026 (5.576)	−15.303 (1.562)	−16.020 (1.515)
log(distance to DC)	6.28	−0.496 (0.269)	−0.006 (0.152)	−0.353 (0.045)	−0.335 (0.043)
Longitude	−88.66	3.048 (2.295)	1.708 (2.791)	7.355 (0.678)	6.697 (0.643)
Latitude	39.54	−0.182 (0.584)	0.713 (0.603)	1.992 (0.187)	1.775 (0.174)
US South	0.29	0.028 (0.087)	−0.096 (0.069)	−0.265 (0.023)	−0.236 (0.021)
Western Union branch	0.78	0.070 (0.010)	0.174 (0.014)	0.349 (0.020)	0.321 (0.017)
Total number of post offices	1,738	593	1,589	1,454	1,763
- of which with treatment:	0	23	29	547	599

(continued)

civil service–classified earlier tended to have more postal employees and a higher number of delivery errors. Reformed cities also have older post offices and were located closer to Washington, DC. These cities tended to be located in counties with a higher population, greater rates of urbanization, and a larger number of manufacturing establishments. Over time, as the civil service reform expands, these differences become smaller but remain sizable.

III. Civil Service Reform and Government Performance

A. Empirical Strategy

Our objective is to examine how the implementation of federal civil service reform affected post office performance. Using the data introduced in Section II, we answer this question using a “stacked” event study design. This research design considers each reform wave (e.g., the initial 1883 statute or the 1893 expansion via executive order) as a separate subexperiment around which we construct a difference-in-differences estimate using cities affected and unaffected in that year. We then “stack” all individual event-specific difference-in-differences to estimate the pooled effect of the civil service reform across all waves.¹¹ As such, let

¹¹In contrast to traditional two-way fixed effects, this research design makes explicit the comparison groups in each period, avoiding econometric issues that a growing literature has highlighted (de Chaisemartin and D’Haultfœuille 2020; Callaway and Sant’Anna 2021). See the review of Roth et al. (2023) for more details.

TABLE 1—DESCRIPTIVE STATISTICS OF REFORMED AND UNREFORMED CITIES (*continued*)

	Mean control	Difference treatment-control in reform wave			
	(1)	1883 (2)	1884–1892 (3)	1893 (4)	Pooled (5)
<i>Panel B. County level</i>					
log(total population)	10.16	1.781 (0.171)	1.165 (0.116)	0.762 (0.037)	0.855 (0.036)
Labor force participation rate	0.361	0.022 (0.007)	0.037 (0.007)	0.015 (0.004)	0.017 (0.003)
Share urban	0.042	0.648 (0.055)	0.470 (0.045)	0.080 (0.009)	0.144 (0.013)
Share literate	0.857	0.030 (0.011)	0.033 (0.018)	0.077 (0.008)	0.071 (0.007)
Nonwhite share	0.106	−0.014 (0.019)	−0.015 (0.031)	−0.063 (0.009)	−0.056 (0.009)
Foreign-born share	0.133	0.104 (0.022)	0.089 (0.019)	0.063 (0.006)	0.067 (0.005)
log(mean occscore)	2.95	0.154 (0.014)	0.152 (0.015)	0.118 (0.007)	0.123 (0.006)
Number of land grant universities	0.052	−0.010 (0.043)	0.085 (0.063)	0.030 (0.013)	0.032 (0.013)
Manufacturing establishments	238.241	2,113.55 (558.06)	601.22 (211.87)	220.314 (18.26)	365.00 (46.48)
Frontier county	0.085	−0.012 (0.043)	−0.084 (0.005)	−0.092 (0.012)	−0.086 (0.011)
Rail access	0.948	0.059 (0.009)	0.066 (0.009)	0.017 (0.006)	0.024 (0.005)
Existing canals	0.101	0.257 (0.103)	0.129 (0.079)	0.083 (0.016)	0.096 (0.016)
Number of party changes	0.217	0.236 (0.087)	0.041 (0.061)	0.066 (0.015)	0.070 (0.015)
Turnout	0.673	−0.035 (0.043)	−0.085 (0.029)	0.012 (0.007)	0.004 (0.007)
Republican vote share	0.443	−0.030 (0.033)	0.008 (0.025)	0.055 (0.008)	0.049 (0.008)
Democrat vote share	0.427	0.042 (0.034)	0.001 (0.026)	−0.041 (0.008)	−0.035 (0.008)
Total number of counties	1,191	484	1,110	1,048	1,203
- of which with treatment:	0	23	29	438	472

Notes: Column 1 shows the mean for the unreformed (control) cities. Columns 2–4 show the difference between reformed and unreformed cities for each reform wave. Column 5 shows the pooled difference, conditional on reform wave fixed effects. Observation counts (total number of cities for panel A, total number of counties for panel B) report the maximum number of observations. See online Appendix B for a description of the data sources, and online Appendix Table A1 for additional county-level characteristics. Robust standard errors are reported in parentheses.

$j = \{1883, 1893\}$ denote the reform wave, and let k be the years before or after civil service protections come into effect. Since k is centered around the date each reform wave takes effect, negative values are years leading up to a civil service reform event, and $k = 0$ denotes the year of the reform. We restrict the pooled sample to the window for which we have coverage in all subexperiments. For delivery errors, for example, the window covers $k = \{-4, \dots, 8\}$. For city i , reform year j , and the k -th year around the reform, we estimate

$$(1) \quad y_{ijk} = \beta \text{treat}_{ij} \times \text{post}_k + \theta_{ij} + \tau_{jk} + \gamma' \mathbf{x}_{ijk} + \varepsilon_{ijk},$$

where $treat_{ij} = 1$ if city i is reformed in the reform wave j , and 0 otherwise. The variable y_{ijk} is the outcome of interest, such as the (log) delivery errors. The variable $post_k$ is defined as $post_k = \mathbf{1}\{k \geq 0\}$, taking the value 1 post-reform, and 0 before. τ_{jk} are reform-specific year fixed effects, which absorb common temporal shocks and reform effects that applied universally (e.g., the prohibition of assessments). Since cities can serve both in the treatment and control groups (e.g., Alameda, California, is untreated until 1893, serving as a comparison city for cities reformed in 1883), we estimate the city fixed effects θ_{ij} separately for each reform wave.¹² The parameter β is the key estimate of interest, capturing the impact of experiencing the civil service reform relative to control cities that do not change their reform status in reform year j . For causal inference, we require that reformed and unreformed cities evolve along common trends in the absence of the reform.

The main identification concern is that reformed cities are not randomly assigned. As shown in Table 1, reformed cities differ from unreformed ones in a variety of ways. Although the inclusion of city fixed effects helps alleviate concerns over level differences, differences across reformed and unreformed cities could still affect the outcome of interest in a time-varying way. For example, if larger cities are both more likely to be reformed and experience improvements in complementary infrastructure (e.g., the telegraph), a simple difference-in-differences will mistakenly attribute effects of such infrastructure improvements to the reform.

There are several ways in which we can address this issue. First, we can use the high-frequency nature of the delivery error data to distinguish changes at the time of civil reform from slower-moving factors. By examining year-by-year estimates, we can be more confident that we are capturing the effect of the specific institutional change. The availability of annual data is thus a significant advance in our study period since economic outcomes for studies examining this time period are often measured using census data that are commonly collected at decadal frequency. Second, we can leverage institutional knowledge about the assignment rule (based on the post office size threshold) to adopt a control function approach by including the time-interacted baseline number of postal employees in the vector of controls $\mathbf{x}_{ijk}$.¹³ Third, we can extend this approach to flexibly control for a wide range of time-interacted baseline characteristics. We use “post-double-selection” to select the covariates out of a battery of 31 variables (Belloni, Chernozhukov, and Hansen 2014).¹⁴ This approach avoids overfitting and provides a principled approach to covariate selection. Finally, we also include state $\times$ year fixed effects to restrict the comparison to cities within the same state, thus allowing for tighter

¹²In our “stacked” panel, control cities are thus excluded if they become reformed within a given event study j ’s time window.

¹³We refrain from referring to this approach as a regression discontinuity design due to the lack of observations around the 50 employees threshold. Instead, our preferred interpretation considers including time-interacted controls more generally as relying on a conditional independence assumption, whereby—conditional on the time-interacted covariates (and the city and time fixed effects), we isolate the exogenous city-year variation stemming from the rollout of the civil service reform.

¹⁴Table 1, online Appendix Table A1, and online Appendix B report descriptive statistics and the source of the full set of variables.

TABLE 2—CIVIL SERVICE REFORM AND PUBLIC SECTOR PERFORMANCE

	log(number of delivery errors)				
	(1)	(2)	(3)	(4)	(5)
Mean of dependent variable:	4.459	4.459	4.459	4.781	4.302
Reform $\times$ Post	−0.220 (0.058)	−0.194 (0.063)	−0.200 (0.080)	−0.517 (0.239)	−0.181 (0.084)
Reform wave					
$\times$ City FEs	✓	✓	✓	✓	✓
$\times$ Year FEs $\times$ State FEs	✓	✓	✓	✓	✓
$\times$ Year FEs $\times$ Postal employment		✓			
$\times$ Year FEs $\times$ PDS controls			✓	✓	✓
Sample	Full sample			1883	1893
Observations	18,856	18,856	18,856	6,201	12,655

Notes: Relating (log) delivery errors to the civil service reform in a stacked event study design (see description equation (1)). The unit of observation is the reform wave $\times$ city $\times$ year. Reform is a dummy that is 1 if the city was covered by the civil service reform in the reform wave, and 0 otherwise. Post is a dummy that is 1 if the year is after the reform year of interest. Column 2 includes the (time-interacted) total number of postal workers (measured in the initial year) as control variables. Column 3 includes time-interacted controls selected via post-double-selection (PDS) (see Belloni, Chernozhukov, and Hansen 2014). The covariates selected via PDS are reported in online Appendix Table B2. Columns 4–5 restrict the sample to only the 1883 and 1893 reform waves, respectively. Standard errors clustered at the city $\times$ reform-wave level.

treatment-control comparisons. We cluster the standard errors for the error term ε_{ijk} at the ij level, corresponding to the level of treatment assignment.¹⁵

B. Effects on Postal Performance

Effects on Delivery Quality.—Table 2 presents the main performance results. The dependent variable is the (log) number of delivery errors in a city and year.¹⁶ The columns vary the specification and samples to probe the robustness of the results. Overall, reformed cities experience a sizable reduction in delivery errors.

Column 1 first reports the baseline specification for equation (1) relying on a simple set of fixed effects without time-varying controls. On average, reformed cities experience a reduction in delivery errors by 22 percent. In columns 2–3, we include time-interacted controls to assess the extent to which baseline differences across reformed versus unreformed cities drive our results. Since the assignment of cities to civil service reform depended deterministically on the total postal employment within an office, column 2 includes the time-interacted baseline postal employment as a control function.¹⁷ In column 3, we probe this result further by flexibly controlling for all available baseline characteristics using post-double selection for data-driven covariate selection (Belloni, Chernozhukov, and Hansen 2014). Out of the 31 baseline characteristics, this exercise adds 3 additional baseline covariates

¹⁵ Our results also hold when clustering at the city level, as well as the county $\times$ reform-wave level or the county level to account for spatial autocorrelation in the errors (online Appendix Table A2).

¹⁶ We do not have data on mail volumes with comparable coverage. In Section IVA, however, we use data on mail volumes for the subset of cities with free delivery to show that the reform also reduced the delivery error rates.

¹⁷ The results are comparable using log-transformation and more flexible, nonlinear functional forms (e.g., up to fourth degree polynomials).

in addition to postal employment: (log) city population size, the county-level share of employees in education, and the presence of a Western Union office.¹⁸ Not surprisingly given the employment-based assignment threshold, these characteristics capture size-related differences across civil service reformed and unreformed cities. Reassuringly, the inclusion of these additional time-interacted controls leaves the estimates relatively unchanged in terms of sign, magnitude, and statistical significance. Throughout the specifications, the reform magnitude for the combined waves of 1883 and 1893 lies between 19 and 22 percent.

A key assumption for the difference-in-differences design to identify the causal effect of civil service reform on post office performance is that treatment and control cities would have experienced common trends in the absence of the reform (“parallel trends”). While this assumption is not directly testable, we can provide supporting evidence for its validity by investigating the presence of pre-trends. Figure 2 provides visual evidence for the reform’s effects on delivery errors by reporting the estimates of the augmented equation (1)—a flexible version of Table 2, column 3—where β is allowed to vary by each year.¹⁹ As Figure 2 shows, we observe little difference between the trends of reformed and unreformed cities prior to the introduction of the reform.²⁰ After the introduction of postal reform, there is a clear reduction in delivery errors in reformed cities relative to those cities that were not reformed. This gap in delivery errors increases gradually before leveling out.

In Figure 3, we reproduce the visual event study evidence broken down by the two major reform waves, with Table 2, columns 4–5 providing the corresponding point estimates. As the figure shows, the dynamic reform effects look strikingly similar across the two main reform waves, with the 1883 reform less precisely estimated—likely due to the smaller number of reform cities. In line with the scale-up literature (e.g., Muralidharan and Niehaus 2017), we find that the reform effects are larger in the initial 1883 expansion but continue to remain sizable in the subsequent large-scale expansion a decade later. The robustness of our effects by reform wave is reassuring. The fact that the same empirical pattern emerges across two different decades suggests that the reform is not driven by any single historical event that coincided with the reform (e.g., a decline in postal rates in the 1880s that affected larger cities), bolstering the external validity of the findings.²¹ The stable pattern across two reform waves also alleviates concerns about time-varying heterogeneous treatment effects that a newer literature increasingly highlights (Roth et al. 2023). More explicitly, online Appendix Figure A1 compares our “stacked” event study estimates with alternative estimation methods developed for staggered difference-in-differences. The dynamic reform effects are again comparable across approaches.

¹⁸ Online Appendix Table B2 reports the baseline covariates selected through post-double-selection for all subsequent regressions.

¹⁹ We estimate $\log(y_{ijk}) = \sum_{l=-4}^8 \beta_l \text{treat}_{ij} \times \mathbf{1}\{k = l\} + \theta_{ij} + \tau_{jk} + \gamma' \mathbf{x}_{ijk} + \varepsilon_{ijk}$, with the pre-reform year ($k = -1$) as the omitted category.

²⁰ Formally testing for differential pre-trends $\text{treat}_{ij} \times k$ on the subsample prior to the reform yields an estimate of 0.030 (SE: 0.047), with a corresponding p -value of 0.52.

²¹ In online Appendix Table A3, we also show that our results hold when including all reform waves.

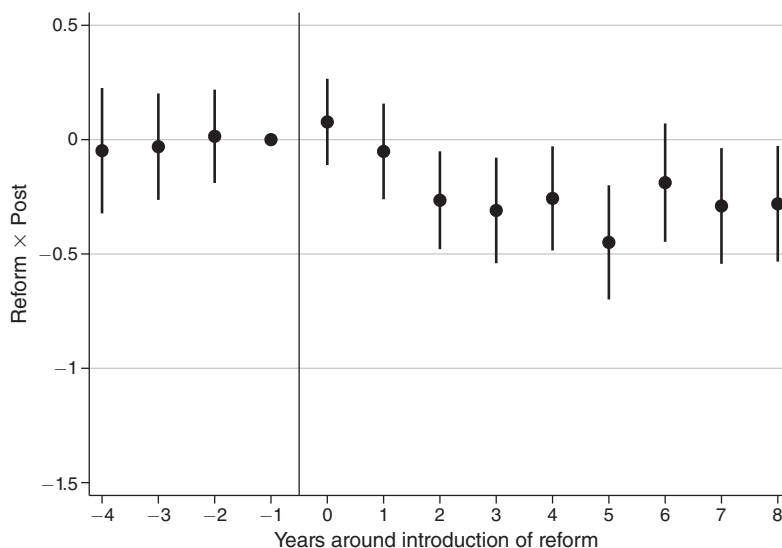


FIGURE 2. DELIVERY ERRORS FOR REFORM VERSUS NONREFORM CITIES AROUND REFORM YEARS

Notes: The figure reports an augmented version of equation (1) (corresponding to Table 2, column 3), where the estimated difference between treatment and control cities is allowed to vary for each year around the introduction of the reform. Reporting 95 percent confidence intervals. Standard errors clustered at the city $\times$ reform year level.

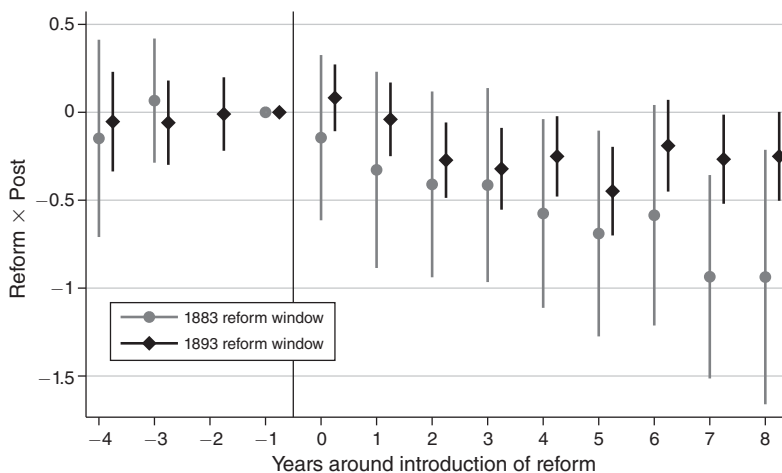


FIGURE 3. DELIVERY ERRORS FOR REFORM VERSUS NONREFORM CITIES BY REFORM WAVE

Notes: The figure reports an augmented version of equation (1) (corresponding to Table 2, column 3), where the estimated differences between treatment and control cities are allowed to vary for each year around the introduction of the reform and are estimated separately for the 1883 and 1893 reform windows. Estimate and confidence interval are missing for relative year $k = -2$ for the 1883 reform wave results due to the absence of an RMS Annual Report for 1881. Reporting 95 percent confidence intervals. Standard errors clustered at the city $\times$ reform year-level.

Effects on Productivity.—Despite the robust evidence of the impact of federal civil service reform on delivery errors, potential concerns may remain over the interpretation of the outcome measure. For example, fewer reports could reflect *worse* performance if errors are not even discovered or if mail is not even routed across

cities. We thus complement our primary outcome measure using the secondary set of productivity measures.

We leverage another source of rich data on the operation of the Post Office Department's City Free Delivery service. These data are available for cities affected by the initial rollout of civil service protections under the 1883 Pendleton Act (see Section IIA). We use two measures of productivity: volume handled per carrier and per unit cost of processing mail. While the free delivery service provides these added outcome measures, a limitation is that they are only available for a smaller subset of 512 larger cities covering the period 1875–1891. As a result, the stacked regression of equation (1) simplifies to a simple difference-in-differences centered around the 1883 reform wave, with errors clustered at the city level. Our main result on delivery errors also holds in this sample, suggesting that zooming in on this subset of the post office to explore mechanisms will still provide generalizable evidence. Given the smaller sample, however, we can no longer include the stringent State $\times$ Year fixed effects, which restrict the empirical analysis to within-state-year comparisons.²²

The regression results are shown in Table 3. In columns 1–2, we focus on volume per carrier as the outcome. Column 1 reports the baseline specification controlling for post office employment. Column 2 reports the specification with the time-interacted controls selected via PDS. Consistent with the improvement in performance shown in Table 2, there is a significant increase in the volume per carrier of 8–14 percent. Columns 3–4 corroborate these results by showing a similar improvement in per unit cost-efficiency (total cost divided by the total volume of mail processed) by 10–13 percent. Column 3 reports the baseline specification, while column 4 reports the version based on PDS. Figure 4 shows the visual evidence. There is a marked relative improvement after the reform but no clear pre-trends. The combined evidence—relying on both delivery errors and productivity measures derived from two independent sources—suggests that civil service reform improved postal performance.

Robustness Checks.—We provide a range of robustness checks to address concerns over more specific confounders. First, a range of infrastructural improvements took place during the nineteenth century that may have shaped information flows, such as the development of the railroads, canals, and the telegraph network. If complementary public infrastructure expanded concurrently with the reform, part of our reform effects may capture general improvements in communication infrastructure. In online Appendix Table A4, we test if civil service reform is associated with the expansion of the telegraph and railroads, the two main ways through which information was transmitted across distance in the nineteenth century. We do so by leveraging complete-count decennial census data to compute the change in telegraph and railroad employment at the county level between 1880 and 1900. As the estimates show, we do not find that the reform is significantly associated with the expansion of the telegraph and railroad, as proxied by employment.

We also investigate confounders related to developments within the post office itself. The implementation of civil service protections within the post office took place against the backdrop of several institutional changes that affected postal service

²² While the sign of the point estimate remains comparable, the inclusion of State $\times$ Year fixed effects results in attenuated and noisier estimates.

TABLE 3—CIVIL SERVICE REFORM AND PRODUCTIVITY (1883 REFORM WAVE)

	log(volume/carrier)		log(cost/volume)	
	(1)	(2)	(3)	(4)
Mean of dependent variable:	12.43	12.43	1.111	1.111
Reform 1883 $\times$ Post	0.084 (0.046)	0.137 (0.068)	−0.100 (0.043)	−0.129 (0.068)
City FEs	✓	✓	✓	✓
Year FEs	✓	✓	✓	✓
Year FEs $\times$ Postal employment	✓		✓	
Year FEs $\times$ PDS controls		✓		✓
Observations	2,880	2,880	2,888	2,888

Notes: Relating (log) total mail volume per carrier (columns 1–2) and (log) total cost per volume (columns 3–4) to the civil service reform in a stacked event study design. The sample is restricted to the 1883 reform wave and covers 1875–1891. The unit of observation is a city $\times$ year. Reform 1883 is a dummy that is 1 if the city was covered by the civil service reform in 1883, and 0 otherwise. Post is a dummy that is 1 if the year is after the reform year of interest. Columns 1 and 3 include the (time-interacted) total number of postal workers as control variables. Columns 2 and 4 include time-interacted controls selected via post-double-selection (see Belloni, Chernozhukov, and Hansen 2014). The covariates selected via PDS are reported in online Appendix Table B2. Standard errors clustered at the city level.

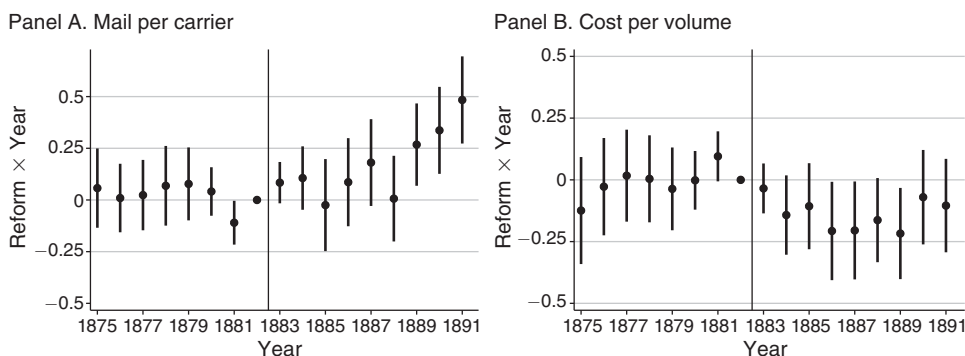


FIGURE 4. DELIVERY PRODUCTIVITY AND COST EFFICIENCY FOR REFORM VERSUS NONREFORM CITIES AROUND THE REFORM

Notes: The figure reports an augmented version of equation (1) (corresponding to Table 3, columns 2 and 4), where the estimated differences between treatment units (reformed cities in 1883) and control units (unreformed cities) are allowed to vary by each year. Panel A shows the event study with (log) overall volume per carrier as the dependent variable, whereas panel B shows the (log) overall cost per volume. Reporting 95 percent confidence intervals. Standard errors clustered at the city level.

during the second half of the nineteenth century. These developments included the advent of free city delivery by carriers in major cities (1863), allowing for money orders to eliminate cash transactions via mail (1864), and the penny postal cards (1873). As the types of services diversified and demand for mail service grew, the post office also increased its internal oversight capacity. Particularly relevant for our study was the official reorganization of postal inspectors under the newly created Division of Postal Inspectors and Mail Depredations in 1875. This period witnessed a modest expansion of staff responsible for carrying out new administrative

responsibilities—many of which were related to a spate of new obscenity laws, beginning with the Comstock Laws of 1873 (Carpenter 2001). Postal inspectors were in turn tasked with identifying fraudulent or obscene material that traveled through the postal system.

Given these institutional changes, one potential empirical concern is that we are confounding the effects of civil service reform with the expanding oversight scope and policing capacity given to postal inspectors in the decade prior to the Pendleton Act. While obscenity laws generally applied nationwide, it is nevertheless possible that postal inspection was targeted at large cities, where vice was viewed as particularly rampant. Given these concerns, we test whether inspection capacity increased within reformed cities after civil service reform took effect (online Appendix Table A5). Reassuringly, we see virtually no effect of civil service reform on the number of inspectors working in affected cities. This finding suggests that the improved postal performance we observe is unlikely to be driven by an expansion in inspection capacity.

Finally, we also consider whether our results are spuriously driven by other concurrent civil service reforms. As Section I describes, the Pendleton Act initially covered substantial parts of the Customs Office in addition to the post office. In online Appendix Table A6, columns 1–3, we show that our reform effects are similar in size and precision when dropping all cities with a customs office, as well as when we drop only the subset of 11 with customs that were reformed. We also consider the possibility that postal reform is associated with concurrent municipal civil service reforms. The federal reform we study predates most of the municipal reforms that occurred in the twentieth century (Rauch 1995; Ornaghi 2019). We show in column 4 that our main results hold when dropping all cities that experience municipal reforms during our study period (online Appendix Table A6). More generally, we demonstrate that the results are not driven by outliers. The main finding holds when dropping one reform city at a time (online Appendix Figure A2) for the 1883 reform wave, as well as when dropping each state at a time for the 1893 reform wave (online Appendix Figure A3). Taken together, the results suggest that the civil service reform within the post office significantly improved post office performance.

IV. Drivers of Increased Performance: Mechanisms

The results above provide robust evidence that postal reform improved public service delivery. As a landmark reform, the Pendleton Act provided a blueprint for many civil service reforms that would follow. Like many civil service reforms today, the Pendleton Act was comprised of distinct provisions, thus creating multiple channels that could affect post office outcomes. We now investigate these potential underlying mechanisms.

A. Changes in Inputs and Workload

Changes in Labor Inputs.—The nineteenth century was a period of significant government growth (Mastrorocco and Teso 2023). As such, we first explore whether the reduction in delivery errors coincides with an expansion of the postal labor force. To test whether our results are associated with an increase in labor inputs,

TABLE 4—POSTAL EMPLOYMENT, DELIVERY ERRORS PER WORKER AND CIVIL SERVICE REFORM

	log(postal staff)		Delivery errors/staff	
	(1)	(2)	(3)	(4)
Mean of dependent variable:	1.548	1.548	0.321	0.321
Reform $\times$ Post	0.014 (0.040)	0.017 (0.041)	−0.081 (0.018)	−0.057 (0.022)
Reform wave $\times$ City FEs	✓	✓	✓	✓
Reform wave $\times$ Year $\times$ State FEs	✓	✓	✓	✓
Reform wave $\times$ Year $\times$ Postal employment	✓		✓	
Reform wave $\times$ Year $\times$ PDS controls		✓		✓
Observations	18,707	18,707	18,707	18,707

Notes: Relating (log) total postal employment (postmaster, clerks, and carriers) and the delivery errors per worker (scaled $\times 100$) to the civil service reform in a stacked event study design. The unit of observation is the reform wave $\times$ city $\times$ year. Reform is a dummy that is 1 if the city was covered by the civil service reform in the reform wave, and 0 otherwise. Post is a dummy that is 1 if the year is after the reform year of interest. Columns 1 and 3 include the (time-interacted) total number of postal workers as control variables. Columns 2 and 4 include time-interacted controls selected via post-double-selection (see Belloni, Chernozhukov, and Hansen 2014). The covariates selected via PDS are reported in online Appendix Table B2. Standard errors clustered at the city $\times$ reform-wave level.

we now make use of the digitized Official Registers, which allow us to track the total employment of mail clerks and letter carriers biennially. We compute the total postal workforce for each city by aggregating the individual-level records. To obtain annual counts, we linearly interpolate between the missing years.

Table 4, columns 1–2 report the estimates based on the total personnel counts. We use the same “stacked” difference-in-differences design (equation (1)), except that the dependent variable now is the (log) total postal employment. There is no strong evidence that the civil service reform significantly affected overall postal employment. The point estimate is positive but quantitatively small, corresponding to an increase in staff of 1.4–1.7 percent, and statistically insignificant. Columns 3–4 further increase our confidence in our main finding by demonstrating that the main performance result holds even when computing delivery error per worker. These results suggest that the improved performance we observe is unlikely to be driven by increasing labor input.

Changes in Worker Quality.—Even if the total number of postal workers remains unaffected, civil service reform may have improved post office performance by improving the quality of serving bureaucrats. Indeed, a large literature has documented the importance of the selection margin for explaining differences in bureaucrat performance (Dal Bó, Finan, and Rossi 2013; Besley et al. 2021). For example, replacing discretionary political appointments with rule-based selection could have attracted higher-quality candidates and thus improved bureaucratic selection. In our context, exam-based selection only applied to new entrants. Replacing the existing personnel was a protracted process,²³ making improved selection *a priori* perhaps a less plausible channel.²⁴

²³ Even 8 years after civil service rules took effect, almost 20 percent of the average reformed post office was still comprised of patronage appointees (online Appendix Figure A4).

²⁴ Implicitly, this argument assumes that workers are substitutes in the production process. To the extent there are strong complementarities, removing “bottlenecks” alone can disproportionately increase postal output.

TABLE 5—CIVIL SERVICE REFORM AND INDIVIDUAL-LEVEL CHARACTERISTICS OF HIRES

<i>Panel A. All hired civil servants</i>	(1)	(2)	(3)	(4)
	Same state	Foreign-born	German	Irish
Mean of dependent variable:	0.628	0.0983	0.0244	0.0187
Reform $\times$ Post	−0.014 (0.014)	0.007 (0.007)	0.007 (0.004)	−0.001 (0.004)
Reform wave $\times$ City FEs	✓	✓	✓	✓
Reform wave $\times$ Year $\times$ State FEs	✓	✓	✓	✓
Reform wave $\times$ Year $\times$ Job FEs	✓	✓	✓	✓
Reform wave $\times$ Year $\times$ Controls	✓	✓	✓	✓
Observations	68,977	68,977	68,977	68,977
<i>Panel B. Census-linked hires</i>	Age	Literacy	Female	White
Mean of dependent variable:	28.40	0.864	0.114	0.966
Reform $\times$ Post	0.826 (1.038)	−0.002 (0.027)	−0.015 (0.026)	−0.000 (0.012)
Reform wave $\times$ City FEs	✓	✓	✓	✓
Reform wave $\times$ Year $\times$ State FEs	✓	✓	✓	✓
Reform wave $\times$ Year $\times$ Job FEs	✓	✓	✓	✓
Reform wave $\times$ Year $\times$ Controls	✓	✓	✓	✓
Observations	22,465	22,465	22,465	22,465

Notes: Relating individual-level characteristics of recruited civil servants to the implementation of the civil service reform. Newly recruited civil servants are identified as workers first observed in the personnel data. To avoid truncation (since all workers are first observed in the earliest year of our data), we exclude the first year of our personnel records, thus covering 1879–1901. The unit of observation is an individual $\times$ reform wave $\times$ city $\times$ year. Reform is a dummy that is 1 if the city was covered by the civil service reform in the reform wave, and 0 otherwise. Post is a dummy that is 1 if the year is after the reform year of interest. Panel A looks at the full sample of recruited civil servants, and panel B looks at the subset of those who could be linked to the US decennial censuses. All specifications include (time-interacted) total postal employment and (log) city population as controls. Standard errors clustered at the city $\times$ reform-wave level.

Nevertheless, we test for the selection channel by analyzing whether the composition of hires changes significantly after the implementation of federal civil service reform. A challenge to the study of bureaucratic selection in the nineteenth century is the lack of data on human capital. We overcome this challenge in several ways. First, we focus on patronage-related measures from the personnel records (the Official Registers). Measures from these records are available for all postal workers, thus eliminating concerns about selection bias. The individual-level results are shown in Table 5, panel A.²⁵ Overall, we do not find significant changes in the composition of the hired postal workforce. Bureaucrats hired post-reform are not differentially likely to be appointed from their home state, nor are they more or less likely to be foreign-born or belong to a nationality associated with significant political machines.

²⁵ Online Appendix Figure A5 reports the corresponding dynamic effects for all individual-level characteristics.

Second, we complement these personnel measures from the Registers by linking these records to the full-count US decennial censuses of 1880 and 1900. This linkage allows us to obtain age (a proxy for previous work experience), literacy (an admittedly crude measure of human capital), and race as additional characteristics. We link our personnel data to the census based on name, birth state, and current state, following Aneja and Xu (2022).²⁶ Table 5, panel B reports the results for the subset of matched civil servants. Reassuringly, we again do not find that civil service reform significantly changed the traits of hired workers. While we find a slight increase in the age and literacy of hired civil servants, the magnitudes are small and statistically insignificant. Similarly, we do not find that the Pendleton Act increased the share of female or nonwhite civil service entrants. In addition to finding little change in worker composition, we also find no evidence of significant changes in the pay of bureaucrats that may indicate improvements in worker quality (online Appendix Table A7).

In sum, the collective body of evidence—the slow rate at which civil servants were hired through the new exam scheme, as well as the absence of large observable changes in the composition of hires—suggests that a change in selection is unlikely to be the main margin of adjustment. While our observable measures of civil servant characteristics are arguably crude given data limitations, many of the point estimates are precisely estimated, allowing us to rule out even relatively small changes in certain individual traits. In Section IVB, we also provide additional evidence using retention by holding constant the selection margin, showing that reform effects applied equally to offers that were selected *prior* to the exams. The absence of uniformly strong effects thus suggests that improved bureaucrat selection is unlikely to be the primary channel through which civil service reform improved the quality of public service delivery offered by American post offices.

Changes in Demand.—Yet another mechanism through which performance might have improved post-reform is through an overall reduction in workload. Even if overall employment levels and the quality of workers remained comparable, the decline in error rates could coincide with a reduction in workload due to a decline in the overall mail volumes. Admittedly, we believe that declining demand for mail services is unlikely to be a confounder in our setting given the increasing scale of postal operations during the nineteenth century (Blevins 2021).²⁷ Nevertheless, we can use data on overall mail volumes from the free delivery statistics for 1875–1891 to test this channel explicitly.²⁸ The results are reported in Table 6, with online Appendix Figure A6 reporting the corresponding dynamic effects. If anything, the results show an *increase* in the overall volume of mail, though the point estimate is

²⁶ We obtain a match rate of 36 percent, which compares favorably to match rates obtained in related work. Importantly, the match rate does not vary significantly with the reform (online Appendix Table A8), and our results are comparable when reweighting the matched sample based on observable characteristics (online Appendix Table C2). Online Appendix C provides additional details on the matching procedure.

²⁷ Moreover, if demand for mail services was increasing during this period, such changes would presumably produce *upward* pressure on delivery mistakes due to increased workload, thus plausibly rendering our effect an underestimate of the true effect of civil service reform.

²⁸ Unfortunately, the free delivery statistics are only reported until 1891. Our delivery results also hold—albeit noisier given the small sample size—for the 1883 expansion and for the subsample of locations for which we have free delivery statistics (online Appendix Table A3).

TABLE 6—TOTAL MAIL VOLUME, COLLECTIONS, DELIVERIES, AND CIVIL SERVICE REFORM (1883 REFORM WAVE)

	log(total volume)		log(collected)	log(delivered)
	(1)	(2)	(3)	(4)
Mean of dependent variable:	14.69	14.69	13.81	14.57
Reform 1883 $\times$ Post	0.122 (0.075)	0.129 (0.130)	0.257 (0.166)	0.104 (0.121)
City FEs	✓	✓	✓	✓
Year FEs	✓	✓	✓	✓
Year FEs $\times$ Postal employment	✓			
Year FEs $\times$ PDS controls		✓	✓	✓
Observations	2,947	2,947	2,947	2,947

Notes: Relating (log) total mail volume (columns 1–2), (log) total mail collected (column 3), and (log) total mail delivered (column 4) to the implementation of the civil service reform in a stacked event study design. The sample is restricted to the 1883 reform wave and covers 1875–1891. The unit of observation is a city $\times$ year. Reform 1883 is a dummy that is 1 if the city was covered by the civil service reform in 1883, and 0 otherwise. Post is a dummy that is 1 if the year is after the reform year of interest. Column 1 includes the (time-interacted) total number of postal workers as control variables. Columns 2–4 include time-interacted controls selected via post-double-selection (see Belloni, Chernozhukov, and Hansen 2014). The covariates selected via PDS are reported in online Appendix Table B2. Standard errors clustered at the city level.

statistically insignificant. In online Appendix Table A9, we disaggregate the overall collection and delivery volumes by type of mail delivered. Once again, the results are inconsistent with the interpretation that the improved performance reflects a decline in delivery volume.

Figure 5 shows the relationship between delivery errors and aggregate postal volume for reformed and unreformed cities. The measures are residualized by partialing out city and year fixed effects, as well as the total postal employment interacted with year fixed effects. Panel A shows the relationship for the subsample of cities that are reformed in 1883, while panel B shows the relationship for the unreformed cities in the same reform wave. As the figure shows, delivery errors tend to increase with higher volumes.²⁹ The introduction of the civil service reform, however, weakened the relationship between errors and volume. Strikingly, we do not observe the same change when examining the relationship in the sample of cities that remain unreformed. This difference in the changing slope across reformed and unreformed cities is statistically significant (online Appendix Table A10). This suggests that civil service reform helped change the “technology” of production, flattening the relationship between delivery errors and overall mail volume.

B. Reduced Turnover and Political Cycles

Prior to the introduction of federal civil service protections, high turnover was a by-product of serving “at the pleasure” of politicians. High turnover was not only a phenomenon around party transitions. As Hoogenboom (1959) explains regarding the pre-reform period, “individuals could anticipate early dismissal from office, for

²⁹The relationship between delivery errors and volumes is flatter among unreformed cities, reflecting the fact that reformed and unreformed cities differ along many dimensions (see Table 1). We focus on the *change in slopes within* reformed and unreformed cities.

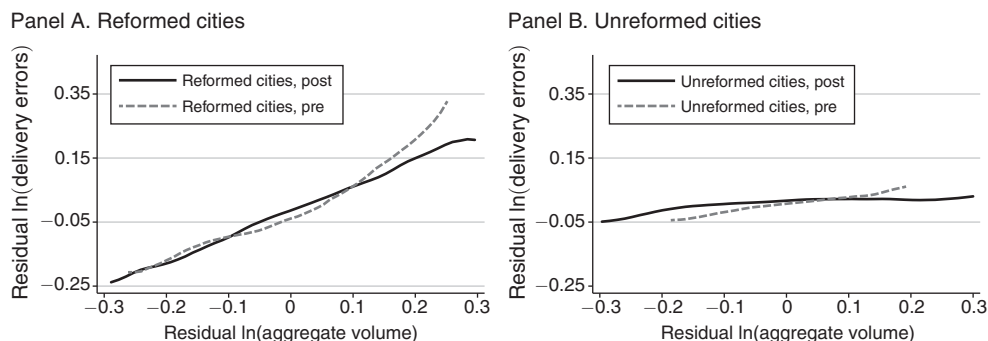


FIGURE 5. DELIVERY ERRORS AND AGGREGATE MAIL VOLUME AROUND THE REFORM

Notes: The figure shows the relationship between (log) residual delivery errors and (log) residual aggregate mail volume (before and after the reform) using a local polynomial. Both variables are residualized by partialling out city and year fixed effects, as well as total employment $\times$ year fixed effects. The sample focuses on the 1883 reform wave, covering cities that are reformed and unreformed between 1875 and 1891. Panel A shows the relationship for cities that are reformed in 1883, and panel B shows the relationship for cities that remain unreformed.

tenure was extremely insecure ... even if his party remained in power, a civil servant was not secure in his position ... [t]hese removals were caused by factional struggles.” The resulting high level of churn not only impeded learning on-the-job but also limited incentives for workers to exert effort or invest in job-specific skills: “morale was low in a civil service largely composed of misfits employed on a temporary basis ... it was impossible for an esprit de corps or loyalty to office or agency to develop in an atmosphere of nervous tension” (Hoogenboom 1959). Motivated by the historical narrative, we directly test whether the Pendleton Act reduced political cycles.

Political Cycles in Exit Rates.—To test whether the civil service reform indeed reduced turnover, we again leverage micro-level personnel data from the Official Registers. We use a “stacked” design analogous to equation (1), except that the outcome now is the exit rate. The results on bureaucratic turnover are reported in Table 7. As the mean of the dependent variable shows, the exit rate in the nineteenth century federal bureaucracy was high: the unconditional probability of exit in a given year was 17 percent. The civil service reform helped significantly reduce the exit rate by 3.7–4 percentage points (columns 1–2). Compared to the mean, this decline is sizable, corresponding to a relative decline of 21–23 percent. Importantly, the decline in exit is driven by presidential electoral cycles. To show this, we restrict the sample to a balanced panel to ensure that the time heterogeneity we uncover is not driven by changes in the composition of cities. As columns 3–4 show, the reform effects hold equally in the balanced panel (column 3) and are driven by exits happening right after a federal election year (column 4). Figure 6 complements the regression results with visual evidence. The figure shows the mean exit rates for reformed versus unreformed cities centered around the first post-reform election.³⁰ Strikingly, the figure shows the presence of significant political cycles

³⁰ For completeness, online Appendix Figure A7 also shows the raw exit rates over time for the two major reform waves of 1883 and 1893.

TABLE 7—CIVIL SERVICE REFORM AND PERSONNEL TURNOVER

	Exit rate				Mean experience	
	(1)	(2)	(3)	(4)	(5)	(6)
Mean of dependent variable:	0.173	0.173	0.148	0.148	1.727	1.727
Reform × Post	−0.037 (0.016)	−0.040 (0.016)	−0.041 (0.023)	0.023 (0.028)	0.967 (0.114)	0.779 (0.138)
Reform × Election				0.054 (0.050)		
Reform × Post × Election				−0.130 (0.053)		
Reform wave × City FEs	✓	✓	✓	✓	✓	✓
Reform wave × Year × State FEs	✓	✓	✓	✓	✓	✓
Reform wave × Year × Postal employment	✓				✓	
Reform wave × Year × PDS controls		✓	✓	✓		✓
Balanced sample			✓	✓		
Observations	9,502	9,502	4,297	4,297	9,232	9,232

Notes: Relating the exit rate (columns 1–4) and mean experience (columns 5–6) to the civil service reform in a stacked event study design. The exit rate is defined as the share of postal workers in a city that is last observed in the data. To avoid truncation (since all workers are last observed in the final year of our data), we exclude the last year of our personnel records, thus covering 1877–1899. The mean experience is the average years postal workers have served in a given post office and year. The unit of observation is a reform wave × city × year. Reform is a dummy that is 1 if the city was covered by the civil service reform in the reform wave, and 0 otherwise. Post is a dummy that is 1 if the year is after the reform year of interest. Columns 1 and 5 include the (time-interacted) total number of postal workers as control variables. Columns 2–4 and 6 include the time-interacted controls selected via post-double-selection (see Belloni, Chernozhukov, and Hansen 2014). The covariates selected via PDS are reported in online Appendix Table B2. Standard errors clustered at the city × reform-wave level.

in the exit rate, consistent with elected politicians exercising their discretion over appointments to restaff the civil service.³¹ These political cycles, however, decline significantly for elections that occur after the civil service reform is implemented.³²

As a natural consequence of the reduced exit rate, we observe a move toward longer and more stable careers. In Table 7, columns 5–6, we relate the civil service reform to the mean years of service, a measure of experience. The results document a significant and economically large improvement in the average tenure. While the mean years of tenure is 1.7 years for the full sample, the average experience in reformed cities increases by 0.78–0.98 years. Overall, our results are thus consistent with the Pendleton Act severing the ties between politics and administration, thereby giving rise to a professionalized service characterized by stable careers.

Bureaucrat Performance during Election Years.—The reduction of turnover and increase in bureaucrat experience is likely to be conducive to greater performance (Akhtari, Moreira, and Trucco 2022). We now directly connect the dampening of political cycles in exit to bureaucrat performance. Similar to the results presented in Table 7, we ask whether the performance improvements are particularly pronounced

³¹These political cycles are also found outside of the post office in other parts of the federal government (see Moreira and Perez forthcoming for evidence in the Customs Office, and Mastroiocco and Teso 2023 for the broader nonpostal federal workforce).

³²Federal employment data are published biennially in odd years, while federal elections occur quadrennially in even years. Our measure of “election year” captures the first observable year after the election, or the year a new presidential term began.

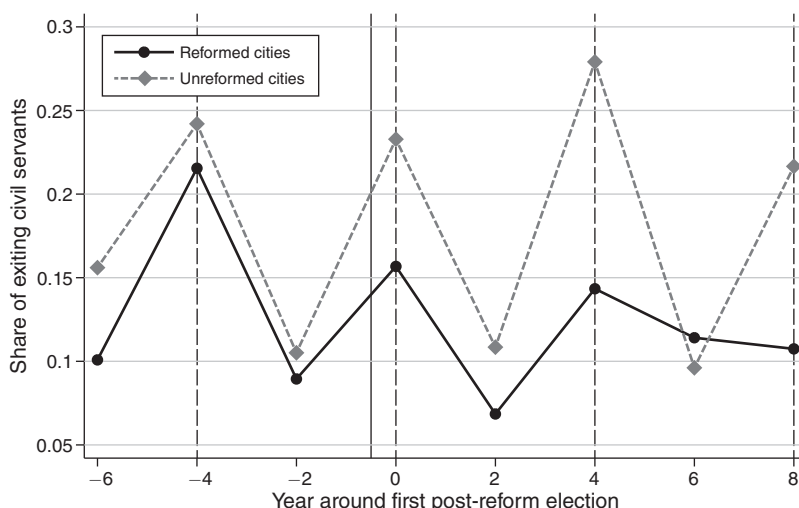


FIGURE 6. EXIT RATES AMONG POSTAL WORKERS IN REFORMED VERSUS UNREFORMED CITIES AROUND THE REFORM

Notes: The figure shows the de-meaned exit rates (i.e., the share of postal workers last observed in a given year) among postal workers in reformed (black solid line) versus unreformed (gray dashed line) cities around the reform. Exit rates are de-meaned within each reform wave $\times$ city. Solid vertical line indicates the year of the reform, and dashed vertical lines indicate presidential election years. Sample includes both major reform waves.

during election years. We restrict the sample to a balanced panel in order to ensure that the time heterogeneity is not driven by composition changes. The results are shown in Table 8. Consistent with the disruptive nature of politically driven turnover, we find that the reduction in delivery errors is driven by election years (column 1). Prior to the reform, these were the years characterized by high turnover. The productivity gains, as measured by greater volume per worker or higher cost-efficiency (columns 2–3), are equally higher in election years. Taken together, these results lead us to conclude that the reduction of patronage-driven turnover was a prominent channel through which civil service reform improved performance.

General versus Exam-Specific Reduction in Exit Rates.—Finally, we probe deeper to understand who is driving the higher post-reform retention. Specifically, we ask whether the reduction in exit applies to all serving officers or is only restricted to those who were recruited through the exam-based selection process. This distinction has implications for the interpretation of our results: if the reform increased retention even for civil servants who were appointed under the spoils system, our results are less likely to reflect changes driven by differential bureaucrat selection. In contrast, if the lower exit rates were only driven by those who were hired through competitive exams, the results would provide a stronger case for the role of bureaucratic selection.

In Table 9, we thus study individual-level heterogeneity. The specification remains the “stacked” event study design, except that the unit of analysis is now the individual-year. Since exit rates vary strongly with tenure, we also introduce reform-specific tenure experience fixed effects. Table 9, column 1 replicates the reduction in exits post-reform at the individual level. In column 2, we restrict the

TABLE 8—PERFORMANCE, ELECTORAL CYCLES, AND CIVIL SERVICE REFORM

	Error rate	Volume/carrier	Cost/volume
	(1)	(2)	(3)
Mean of dependent variable:	0.278	12.59	1.019
Reform $\times$ Post	0.002 (0.034)	−0.003 (0.117)	−0.066 (0.109)
Reform $\times$ Election	0.060 (0.040)	−0.090 (0.038)	0.078 (0.041)
Reform $\times$ Post $\times$ Election	−0.075 (0.041)	0.194 (0.072)	−0.157 (0.073)
Reform wave $\times$ City FEs	✓	✓	✓
Reform wave $\times$ Year $\times$ State FEs	✓		
Reform wave $\times$ Year FEs		✓	✓
Reform wave $\times$ Year $\times$ PDS controls	✓	✓	✓
Balanced panel	✓	✓	✓
Sample	All waves	1883 reform wave	
Observations	9,902	1,085	1,088

Notes: Relating delivery errors (column 1), (log) overall mail volume per carrier (column 2), and (log) cost per volume (column 3) to the civil service reform in a stacked event study design. The unit of observation is a reform wave $\times$ city $\times$ year. Reform is a dummy that is 1 if the city was covered by the civil service reform in the reform wave, and 0 otherwise. Post is a dummy that is 1 if the year is after the reform year of interest. Election is a dummy that is 1 in a presidential election year, and 0 otherwise. Column 1 covers all reform waves, and columns 2–3 restrict the sample to only the 1883 reform wave. All specifications include time-interacted controls selected via post-double-selection (see Belloni, Chernozhukov, and Hansen 2014). The covariates selected via PDS are reported in online Appendix Table B2. Standard errors clustered at the city $\times$ reform-wave level.

sample to only individuals who were *already* serving when the reform was implemented. Holding selection constant, the results suggest that the retention effects even accrue to those who were appointed by patronage.³³ In column 3, we restrict the sample to only the post-reform period, when we observe the exit rates for those entering before and after the reform operating in the same incentive environment. As the point estimate shows, we do not find that the exit rates are significantly different for those who entered after the competitive exams were introduced. In column 4, we include both regressors jointly to decompose the general retention effect from the cohort-specific selection effect. The results suggest that the improvement in retention (i.e., lower exit) holds for all civil servants, irrespective of their entry cohort. This strongly suggests that the primary channel through which the civil service reform worked was by limiting arbitrary firing, as opposed to improving selection.

V. Civil Service Reform and the Weakening of Local Political Parties

A central motivation for civil service reform was to reduce the power of local parties over the hiring of government employees (Mashaw 2009). While the power of appointments rested de jure with the federal government, hiring decisions were de facto delegated to the local level during the patronage era, due to the scale

³³For completeness, online Appendix Figure A9 reports the corresponding dynamic effects.

TABLE 9—CIVIL SERVICE REFORM AND EXIT, BROKEN DOWN BY THE TIME OF HIRING

	Exit			
	(1)	(2)	(3)	(4)
Mean of dependent variable:	0.121	0.146	0.124	0.121
Reform $\times$ Post	-0.032 (0.009)	-0.160 (0.022)		-0.053 (0.014)
Reform $\times$ Post-reform entry cohort			0.014 (0.014)	0.026 (0.014)
Reform wave $\times$ City FEs	✓	✓	✓	✓
Reform wave $\times$ Experience FEs	✓	✓	✓	✓
Reform wave $\times$ Year $\times$ Job FEs	✓	✓	✓	✓
Reform wave $\times$ Year $\times$ State FEs	✓	✓	✓	✓
Reform wave $\times$ Year $\times$ Controls	✓	✓	✓	✓
Sample	Full sample	Pre-reform entrants	Post-reform period	Full sample
Observations	143,754	68,853	104,938	143,754

Notes: Relating individual-level exit to the civil service reform in a stacked event study design. The unit of observation is an individual $\times$ reform wave $\times$ city $\times$ year. Reform is a dummy that is 1 if the city was covered by the civil service reform in the reform wave, and 0 otherwise. Post is a dummy that is 1 if the year is after the reform year of interest. Post-reform entry cohort is a dummy that is 1 if the individual entered (i.e., was first observed) after the implementation of the civil service reform. Columns 1 and 4 report results based on the full sample. Columns 2–3 split the sample by whether individuals entered before (Post-reform entry cohort = 0) or after the reform (Post-reform entry cohort = 1). All specifications include (time-interacted) total postal employment and (log) city population as controls. Standard errors clustered at the city $\times$ reform-wave level.

of the government and costs associated with centralized monitoring by federal officials (Section I). This gave influential local politicians substantial political rents (Hoogenboom 1959). By limiting the scope for political interference and politically motivated personnel decisions, civil service reform likely reduced the power of local party machines, which depended on patronage for political and financial support. By shielding rank and file civil servants from political interference, it is possible that the Pendleton Act not only improved postal performance but also helped weaken the power of local political parties.

Unfortunately, a major constraint to studying the downstream political effects of the Pendleton Act is the lack of data on either city-level elections or local party activities during our study period. Guided by research on the history of American political development, we use the local presence and circulation of explicitly partisan newspapers as a proxy for the strength of partisan politics. During the nineteenth century, the newspaper was not only the main source of news but was also central to political parties' local-level organizational activity (Baldasty and Rutenbeck 1988). Most papers were associated with a particular party, with the editor often being considered a party officer. Local parties supplied readership (requiring members to subscribe), and newspapers promoted party ideas and candidates (Kaplan 2002; Baldasty and Rutenbeck 1988). Political parties also supported newspapers financially, with parties funding explicitly partisan newspapers (Kaplan 2002).

While the time period during which the federal government adopted civil service reforms coincided with the well-noted decline in the number of explicitly partisan newspapers (Petrova 2011; Gentzkow et al. 2015), how civil service reform impacts

the presence of partisan newspapers is, in theory, ambiguous. On the one hand, given that civil service reform reduced the power of party elites, it is reasonable to postulate that civil service reform also had potential downstream effects on agents supported by parties—such as the partisan press. With fewer patronage jobs to offer, local parties lost membership—which in turn may have reduced the readership of party-controlled newspapers. Without the support of a strong membership base or political assessments, parties were potentially less able to support the partisan press financially, thus leading to a *decrease* in the presence of political newspapers.³⁴ On the other hand, civil service reform may also *increase* the importance—and thus improved the viability—of partisan newspapers. Patronage jobs for party loyalists (and party newspaper readers) were not the only form of patronage that officials controlled during the nineteenth century. Newspaper editors also received direct support from politicians in the form of government printing contracts and other forms of subsidies (Dyer 1980). As such, to the extent that political parties needed party-run newspapers to mobilize the electorate in support of a candidate—and were no longer able to secure votes by the promise of government positions—one might reasonably expect them to use other methods such as government subsidies to preserve the strength of partisan newspapers. Despite the importance of the Pendleton Act and the rise of nonpartisan media in US economic history, however, there has been to this point little work that aims to quantitatively establish the link between civil service reform and the proliferation of partisan media.

To test the hypothesis that civil service reform weakened the role of local politics, we use city-level newspaper data from Gentzkow, Shapiro, and Sinkinson (2011). The data provide lists of newspapers along with their circulation and partisanship for each city and election-year (see Section II for a detailed description of the data). We collapse the data to compute the total number of newspapers and circulation for each city and election-year, broken down by partisanship. We then adopt the same “stacked” event study design to ask if cities affected by civil service reform also exhibit changes in the number of partisan and nonpartisan newspapers and their circulation.

Table 10 presents the results.³⁵ The dependent variable in columns 1–2 measures the total number of newspapers that are political or independent. The number of explicitly partisan newspapers declines significantly after the reform (column 1).³⁶ While the average number of political newspapers in a city-year is 1.8, the number declines by 0.2 in reformed cities, reflecting a relative decline by 11 percent. In contrast, we observe an increase in independent newspapers (column 2). Columns 3–5 focus on circulation for cities where local newspapers are present. Consistent with the extensive margin effects, civil service reform decreases the circulation share of

³⁴The channel we propose—which links the removal of patronage jobs to fewer political rents—is not exclusively limited to post office employment. The only other federal organization that was affected by civil service protections in the act’s initial years was the Treasury’s custom houses (Hooogenboom 1959). Among those 11 cities with reformed customs, all also saw their post office reformed, thus making it impossible to separately estimate the effect of the customs reform on the presence of political newspapers.

³⁵Since newspaper data are only available for election years for the period 1879–1900, the number of observations is somewhat smaller.

³⁶Online Appendix Figure A10 shows the visual evidence, centering elections around the reform. A limitation of the data from Gentzkow, Shapiro, and Sinkinson (2011) is that we only have a single pre-period. Nonetheless, the figure suggests a sharp decline and subsequent leveling out.

TABLE 10—CIVIL SERVICE REFORM AND THE DECLINE IN LOCAL PARTISAN NEWSPAPERS

	Newspaper number		Circulation		Circ. share political
	Political	Independent	Political	Independent	
	(1)	(2)	(3)	(4)	
Mean of dependent variable:	1.821	0.191	5.578	0.834	0.907
Reform × Post	−0.195 (0.079)	0.085 (0.043)	−0.266 (0.779)	0.800 (0.477)	−0.044 (0.025)
Sample	Election years				
Reform waves	All reform waves				
Data	Gentzkow et al. (2011)				
Reform wave × City FEs	✓	✓	✓	✓	✓
Reform wave × State × Year FEs	✓	✓	✓	✓	✓
Reform wave × Year × PDS controls	✓	✓	✓	✓	✓
Observations	4,235	4,235	3,758	3,758	3,758

Notes: Relating downstream effects to the civil service reform in a stacked event study design. In columns 1–2, the dependent variable is the total number of newspapers in a given city and year, broken down by whether they are political (i.e., affiliated with a political party) or not (independent). In columns 3–4, the dependent variable is the corresponding number of newspaper circulation (in 1,000s) that is political or independent. Column 5 is the share of newspapers in circulation that is political. All specifications include the time-interacted controls selected via post-double-selection (see Belloni, Chernozhukov, and Hansen 2014). The covariates selected via PDS are reported in online Appendix Table B2. Standard errors clustered at the city × reform-wave level.

political newspapers, while significantly increasing the share of independent newspapers in circulation (columns 3–4). Overall, this results in a relative decline in the share of political newspapers in circulation (column 5). While the intensive margin estimates are less precisely estimated, the combined evidence suggests that civil service reform indeed helped increase the presence and circulation of independent media.

VI. Conclusion

Institutions that protect civil servants from political interference are nowadays considered defining features of a well-functioning state. While political appointees make up less than 1 percent of the US federal civil service today (Spenkuch, Teso, and Xu 2023), this was not always the case. Prior to the introduction of the Pendleton Act of 1883, virtually all federal workers were politically appointed. In this paper, we studied the first major expansion of civil service protections in the United States. We show that a strengthening of state capacity through civil service reform improved the quality of a major government institution: the United States Postal Service. Arguably, there are few government institutions that have been as central to modern state building as the post office (Gallagher 2016; John 1995). Postal expansion in Western Europe and the United States in the nineteenth century produced “greater strides in the improvement of communication than had taken place in all previous centuries” (Howe 2007, p. 5). For most Americans in the nineteenth and early twentieth centuries, “the postal system was the central government” (John 1995). By facilitating the flow of information and knowledge through the mail, the post office connected people across a vast and expanding nation.

We leverage the gradual rollout of federal civil service reform across cities and rich data on delivery errors to demonstrate that the Pendleton Act—a hallmark civil

service reform—indeed led to a strengthening of state capacity. Cities that were covered by the civil service reform saw a significant reduction in delivery errors. Making use of personnel data and rich statistics on city free delivery, we show that the reduction in delivery errors likely reflected an increase in productivity. Finally, we use newly digitized personnel records of postal workers to open the black box of the organization and shed light on the mechanisms. Consistent with the historical literature (Johnson and Libecap 1994; Hoogenboom 1959), we find that the civil service reform led to a significant reduction in turnover and more stable careers.

As a key infrastructure, the functioning of the postal service is likely to have broader aggregate impacts (Gentzkow, Shapiro, and Sinkinson 2011). We provide suggestive evidence that a depoliticized and more efficient postal service helped pave the way for the rise of independent newspapers. Leveraging data on newspaper partisanship and circulation, we find that the civil service reform significantly weakened the presence of partisan newspapers. In a context where political newspapers served at the behest of local party leaders (Petrova 2011), our results thus indicate the role of civil service reform in weakening the power of local parties. Our findings thus suggest a “double-dividend” of civil service reform: separating politics from administration in a rank and file bureaucracy not only improved public sector performance but also helped weaken the role of local partisanship.

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